Bhim Rajbhar

Data Scientist | Machine Learning Intern

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github.com/BhimAIEngineer   kaggle kaggle.com/bhimaiengineer   leetcode.com/u/bhim-rajbhar

PROFESSIONAL SUMMARY

AI/ML Engineer with 3.8 plus years designing and deploying machine learning, computer vision, and agentic AI systems across the full lifecycle. Hands-on experience building multi-agent systems with LangGraph, including intent classification, rule-based task routing, and planner-style orchestration across specialist agents, plus Retrieval-Augmented Generation (RAG) pipelines backed by a FAISS vector database and rule-based guardrails for safety, compliance, and PII redaction. Proficient in Python for backend AI systems and RESTful microservices using FastAPI, with production deployment experience using Docker, Kubernetes, CI/CD, and MLflow across AWS, Microsoft Azure, and Google Cloud Platform. Strong mathematical and statistical foundation from production-scale data analysis at a Fortune 500 retail account. Google Cloud Certified Professional Machine Learning Engineer and Microsoft Certified Azure Data Scientist Associate. Currently pursuing a Master of Computer Applications with specialization in AI and ML. Comfortable working cross-functionally to translate business needs into deployed solutions and to present model trade-offs to technical and non-technical stakeholders.

EXPERIENCE

Data Scientist — 12/2021 – 07/2025 | Mumbai, India

Tata Consultancy Services Limited

- Designed, trained, and evaluated machine learning and deep learning models, including classification, regression, and computer vision (OpenCV, YOLO), to automate visual quality-inspection steps for a Fortune 500 retail client's supply chain, previously performed manually.
- Prepared and pre-processed large-scale transactional and customer browsing datasets using Python, NumPy, and Pandas, including data cleaning, normalization, feature extraction, and transformation to ready data for model training and testing.
- Trained and optimized classification and regression models with Scikit-learn for store-level inventory replenishment, selecting algorithms and tuning hyperparameters through cross-validation, and evaluating performance using accuracy, precision, recall, and F1-score.
- Built time-series forecasting models (ARIMA, LSTM, Prophet) for demand prediction across multiple stages of the retail supply chain, and applied hypothesis testing and statistical distribution analysis to validate genuine sales shifts before acting on them.
- Deployed machine learning and computer vision models to production using Docker and FastAPI within CI/CD pipelines; implemented Prometheus-based drift monitoring that caught gradual model degradation after a holiday-driven shift in regional buying patterns.
- Wrote complex SQL queries against the client's PostgreSQL data warehouse to extract store-level SKU, supplier, and inventory data, keeping heavy aggregation in-database rather than loading large tables into memory.
- Tracked experiments and maintained model versioning using MLflow and Weights and Biases; used Git and GitHub for version control and Kubernetes for workload orchestration across AWS, Azure, and Google Cloud Platform.
- Collaborated cross-functionally with data engineers and business stakeholders to translate requirements into deployed solutions, and prepared presentations and reports communicating model results and trade-offs to non-technical audiences.

Machine Learning Intern — 06/2026 | Remote

SmartED Innovations

- Contributing to machine learning workflows for an EdTech e-learning platform, gaining hands-on experience across the ML lifecycle from data preprocessing through model experimentation.
- Using Python, Pandas, and Scikit-learn for data cleaning and pipeline preparation, with growing hands-on exposure to model development using TensorFlow and PyTorch.
- Exploring Generative AI and LLM-based agent features using LangChain and the OpenAI API, supporting early-stage prototyping of AI-driven functionality.
- Collaborating with the engineering team to research and evaluate ML and AI approaches for education-technology use cases.

PROJECTS

Multi-Agent Healthcare Triage/Support Assistant : — 05/2026 – 07/2026

Python, LangGraph, FastAPI, FAISS, Docker

- Designed and built a multi-agent triage assistant using LangGraph, implementing intent classification, rule-based task routing, and planner-style orchestration across three specialist agents: a Retrieval-Augmented Generation (RAG) question-answering agent, a tool-calling scheduling agent, and an emergency-escalation agent.
- Implemented a FAISS vector database-backed RAG pipeline alongside a rule-based guardrail layer for diagnosis-language mitigation and PII redaction, enforcing safety and compliance boundaries before responses reach the user.
- Exposed the multi-agent pipeline as a FastAPI microservice with an interactive Streamlit dashboard, containerized with Docker, and validated by a 14-test suite run through GitHub Actions CI/CD.

CORE COMPETENCIES

Python, LangGraph, LangChain, Agent Development, Multi-Agent Orchestration, Prompt Engineering, LLM APIs (OpenAI, Anthropic), Retrieval-Augmented Generation (RAG), Vector Databases (FAISS), Guardrails and PII Redaction, Responsible AI, FastAPI, REST APIs, Microservices, Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, AWS, Microsoft Azure, Google Cloud Platform, Machine Learning, Deep Learning, Computer Vision (YOLOv8, OpenCV), Natural Language Processing, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, SQL, Time-Series Forecasting, Data Analysis, Git and GitHub.

SKILLS

Agentic AI and LLM Engineering :
Python (backend AI systems, microservices) - Expert; LLM APIs (OpenAI, Anthropic); Prompt Engineering; Agent Development (LangGraph, LangChain); Multi-agent orchestration (intent classification, task routing, planner-style patterns); Retrieval-Augmented Generation (RAG) pipelines; Vector databases (FAISS); Guardrails and safety (PII redaction, diagnosis-language mitigation); Responsible AI (bias and fairness evaluation).

Computer Vision :
OpenCV, YOLO (YOLOv8), CNN architectures (ResNet50), object detection, image classification, image processing.

NLP and Generative AI :
NLP (BERT, Hugging Face Transformers), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, OpenAI API.

Data Preparation and Analytics :
Data cleaning, normalization, feature extraction and transformation, exploratory data analysis, statistical hypothesis testing, SQL (PostgreSQL), NoSQL.

Version Control and Collaboration :
Git and GitHub; cross-functional delivery with data engineers and business stakeholders; presenting technical findings and model trade-offs to non-technical stakeholders.

Machine Learning and Modeling :
Linear regression, decision trees, random forests, support vector machines, deep learning; classification, regression, clustering, time-series forecasting (ARIMA, LSTM, Prophet); hyperparameter tuning, cross-validation, model evaluation (accuracy, precision, recall, F1-score).

Production Deployment and MLOps :
Docker, FastAPI, Flask, REST APIs, Kubernetes, CI/CD (GitHub Actions), MLflow, Weights and Biases, DVC, Prometheus-based model monitoring and drift detection.

Frameworks and Libraries :
Python (Expert), TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, OpenCV, Pillow, Hugging Face Transformers, Sentence Transformers, LangChain, LangGraph, LlamaIndex, FAISS, ChromaDB, Pinecone, Weaviate, XGBoost, LightGBM, CatBoost, NLTK, spaCy, FastAPI, Flask, Streamlit, MLflow, ONNX, TensorBoard, Gradio.

Cloud Platforms :
AWS, Microsoft Azure (Azure ML), Google Cloud Platform - model training and deployment.

Soft Skills — Expert
Stakeholder management, requirement gathering, attention to detail, project ownership, cross-functional collaboration.

CERTIFICATES

Google Certified: Professional Machine Learning Engineer Certification.
Google Cloud Certified: Professional Machine Learning Engineer. Certified in building, evaluating, and productionizing machine learning models on Google Cloud Platform, including responsible AI practices, model fairness evaluation, and workflows for large-scale datasets.

Microsoft Certified: Azure Data Scientist Associate
Microsoft Certified: Azure Data Scientist Associate. Certified in designing machine learning solutions, exploring data and running experiments, training and deploying models, and optimizing language models for AI applications on Microsoft Azure.

Microsoft Certified: Power BI Data Analyst Associate
Microsoft Certified: Power BI Data Analyst Associate. Certified in data preparation, data modeling, data visualization and analysis, and Power BI administration and security.

EDUCATION

Master of Computer Applications With Specialization in AI / ML — 08/2025 – Present | Dehradun, India
Uttaranchal University
Relevant coursework: Python for Data Science, Data Structures, Design and Analysis of Algorithms, Machine Learning, Information Retrieval, Artificial Intelligence, Deep Learning, Natural Language Processing, Data visualization, Business Case Studies.

Bachelor of Science in Information Technology — 06/2018 – 07/2021 | Mumbai, India
University of Mumbai

HSC - Computer Science (Higher Secondary Certificate - Computer Science) — 06/2016 – 03/2018 | Mumbai, India
Maharashtra State Board

Declaration

I am confirming that all the information provided above is accurate to the best of my knowledge.

Bhim Rajbhar